Avijit Chowdhury — Curriculum Vitæ

Post-Doctoral Fellow,Indian Institute of Astrophysics,Bengaluru, India.
(+91) 7908546565 • chowdhury.avijit.001@gmail.com

Personal Data

Date of Birth: 25 January 1992

Permanent Address: 20, Kapali Para, 1st Lane, Budge Budge, Kolkata, West Bengal, India, Pin: 700137

Funding, Fellowship, Grants

Anusandhan National Research Foundation, Department of Science and Technology, Govt. of India
International Travel Support (To attend GR24&Amaldi16 conference in Glasgow,UK) *June 2025*

Anusandhan National Research Foundation, Govt. of India
National Post Doctoral Fellowship *April 2024 – March 2026*

Institute for Nuclear Sciences, National Autonomous University of Mexico (UNAM)
FORDECYT-PRONACES- CONACYT postdoctoral grant, Mexico *August 2021 (not availed)*

Experience

Indian Institute of Astrophysics
Post Doctoral Fellow, Bengaluru, India *March 2026 – Present*

Indian Institute of Technology Guwahati
National Post Doctoral Fellow, Guwahati, Assam, India *April 2024 – February 2026*

Indian Association for the Cultivation of Science
Post Doctoral Fellow *June 2023 – March 2024*

Indian Institute of Technology Bombay
Post Doctoral Fellow *June 2021–June 2023*

Education

Indian Institute of Science Education and Research Kolkata
Ph.D. (Integrated Ph.D. Program) in Physics *2015–2021*
Advisor: Prof. Narayan Banerjee
Thesis Title: On the effect of scalar fields on Hawking radiation and quasinormal modes of black holes

Indian Institute of Science Education and Research Kolkata
Masters in Physics (Integrated Ph.D. Program) *2013–2015*
Advisor: Prof. Narayan Banerjee
Thesis Title: On the Laws of Black Hole Mechanics for Hairy Black Holes

Asutosh College, University of Calcutta
Bachelor of Science(Hons.) in Physics *2010–2013*
Kolkata, India

M. P. Birla Foundation Higher Secondary School
Indian School Certificate Examination *2008–2010*
Kolkata, India

St. Stephen's School
Indian Certificate of Secondary Education *2008*
Kolkata, India

Academic Membership

- Life member of Indian Association for General Relativity and Gravitation (IAGRG).

Academic Responsibilities

2022–Present: Reviewer for international scientific journals: Physical Review Letters, Physical Review D, The European Physical Journal C, General Relativity and Gravitation

Research summary

I work in theoretical gravitational physics, using black holes as natural laboratories to test our understanding of gravity under the most extreme conditions in the universe. My research takes a multi messenger perspective, using different kinds of observations as complementary views of the same object. When black holes are disturbed, for example when two of them merge, they emit gravitational waves that can be detected by observatories such as LIGO. At the same time, the hot gas surrounding them can produce light that telescopes can observe. I study how a black hole's environment, including dark matter and infalling matter, subtly shapes both kinds of signals. By combining what gravitational waves and light reveal, we can develop more complete and reliable tests of whether an observed object is truly a black hole, as predicted by Einstein's theory, or something more exotic. My work connects mathematical theory and computational modelling with observations from current and future gravitational wave detectors and telescopes.

Publications

Articles published in conference proceedings

1. Dark matter halos and transonic accretion flow, Accepted for publication in *Journal of Physics: Conference Series* in the proceedings of *The 24th International Conference on General Relativity and Gravitation (GR24) and the 16th Edoardo Amaldi Conference on Gravitational Waves (Amaldi16)*, J.Phys.Conf.Ser. 3177 (2026) 1, 012038

Journal publications

12. A. Bhowmik, **A. Chowdhury** and S. Chakrabarti "The Ringdown and the Tide: Fingerprints of Dark Matter Halo Profiles", arXiv: 2608.07678 [gr-qc], Accepted in PRD
11. **A. Chowdhury**, G. Sen, S. Chakrabarti,S. Das "Effect of generic dark matter halo on transonic accretion onto galactic black holes", Phys.Rev.D 112 (2025) 6, 064041.
10. R. Ghosh, A. K. Mishra and **A. Chowdhury**, "Birkhoff's Theorem and Uniqueness: A peek beyond General Relativity", Phys.Lett.B 871 (2025) 139974.
9. **A. Chowdhury**,A. K. Mishra and S. Chakraborty, "Connecting quasi-normal modes with causality in Lovelock theories of gravity", Phys.Lett.B 861 (2025) 139219.
8. **A. Chowdhury**, S. Biswas and S. Chakraborty, "Accreting Schwarzschild-like compact object: Plasma-photon interaction and stability", Phys. Rev. D **110**,(2023) 064072 .
7. **A. Chowdhury**, S. Xavier and S. Shankaranarayanan, "The dominating mode of two competing massive modes of quadratic gravity", Sci. Rep. **13**, 8547 (2023).
6. **A. Chowdhury**, S. Devi S. Chakrabarti, "Naked singularity in 4D Einstein-Gauss-Bonnet novel gravity: Echoes and instability", Phys. Rev. D **106**, (2022), 024023.
5. **A. Chowdhury** and N. Banerjee, "Echoes from a singularity", Phys. Rev. D **102**, (2020), 124051.
4. **A. Chowdhury** and N. Banerjee, "Greybody factor and sparsity of Hawking radiation from a charged spherical black hole with scalar hair", Phys. Lett. B **805**, (2020), 135417.
3. **A. Chowdhury**, "Hawking emission of charged particles from an electrically charged spherical black hole with scalar hair", Eur. Phys. J. C **79**, 928 (2019).
2. **A. Chowdhury** and N. Banerjee, "Superradiant stability of mutated Reissner–Nordström black holes", Gen. Rel. Grav. **51**, 99 (2019).
1. **A. Chowdhury** and N. Banerjee, "Quasinormal modes of a charged spherical black hole with scalar hair for scalar and Dirac perturbations", Eur. Phys. J. C **78**, 594 (2018).

Communicated Preprints

2. R. Ghosh, N. Speeney, **A. Chowdhury**, C. Singha, M. A. S. Pinto and E. Berti, "From Density to Mass: A New Parametrized Framework for Dark Matter Environment Around Black Holes",arXiv: 2609.35771 [gr-qc]
1. **A. Chowdhury**, C. Singha, K. Bamba, S. Chakraborty "Tidal deformation of an accreting compact object", arXiv: 2607.24938 [gr-qc].

Link to my INSPIRE-HEP, Google Scholar,ORCiD

Scientific Presentations

Invited

June 10 – 12, 2026: Poster Presentation on *"Probing Dark Matter Around Black Holes: Multi-Messenger Signatures from Black Hole Accretion and Ringdown Spectroscopy"* at the **Symphony of spacetime** organized by the Raman Research Institute, Bangalore

October 6 – 10, 2025: Oral Presentation on *"Dark matter halo and transonic accretion flow"* at the **10 Years of Gravitational Wave Discovery: Advancing the Frontiers of Gravitational Physics** organized by the Indian Institute of Technology Gandhinagar

July 15, 2025: Oral Presentation on *"Effect of generic dark matter halo on transonic accretion onto galactic black holes Environmental Influences on Compact Objects: Plasma Effects and Dark Matter Halos"* at the **School of Physics and Astronomy, University of St. Andrews, Scotland, UK**

July 22–24, 2024: Oral Presentation on *"EFT approach to QNMs in Modified Gravity"* at the **Testing gravity with multimessenger astronomy** hosted by the Indian Institute of Technology Bombay, India.

February 5–10, 2024: One-day lecture on *"Black hole perturbations, Quasinormal modes and quasibound states"* at the Department of Physics, National Institute of Technology, Rourkela, India (NIT-R)

May 26, 2022: Oral Presentation on *"Perturbation of spacetimes with naked singularity"* at Instituto de Ciencias Nucleares, Universidad Nacional Autónoma de México, Mexico City, Mexico.

December 11, 2020: Oral Presentation on *"Scalar fields, Quasinormal modes and echoes"* at the Centre for Theoretical Studies, Indian Institute of Technology Kharagpur, India.

Contributed

May 15–19, 2026: Poster Presentation on *"Effect of generic dark matter halo on transonic accretion onto galactic black holes"* at the **44th meeting of the Astronomical Society of India** hosted by the Indian Institute of Technology Guwahati, India

July 14–18, 2025: Oral Presentation on *"Effect of generic dark matter halo on transonic accretion onto galactic black holes"* at the **24th International Conference on General Relativity and Gravitation and 16th Edoardo Amaldi Conference on Gravitational Waves** hosted by the Institute for Gravitational Research at the University of Glasgow and the Institute of Physics., Glasgow, UK.

January 02–04, 2025: Oral Presentation on *"Accreting Schwarzschild-like compact object: Plasma-photon interaction and stability"* at the **33rd Meeting of the Indian Association for General Relativity and Gravitation** hosted by the Birla Institute of Technology and Science, Pilani, India.

March 01–05, 2023: Oral Presentation on *"How a non-detectable mode can provide a hint of modified gravity?"* at the **41st meeting of the Astronomical Society of India** hosted by the Indian Institute of Technology Indore, India.

December 19–21, 2022: Oral + Poster Presentation on *"Massive spin-2 mode carries more energy than spin-0 mode in quadratic gravity"* at the **32nd meeting of the Indian Association for General Relativity and Gravitation** hosted by the Indian Institute of Science Education and Research Kolkata, India.

July 5, 2022: Oral Presentation on *"Massive tensor modes carry more energy than scalar modes in quadratic gravity"* at the **23rd International Conference on General Relativity and Gravitation** organized by the Institute of Theoretical Physics, Chinese Academy of Sciences, China.

January 19–20, 2020: Oral Presentation on *"Echoes from a singularity"* at the **31st meeting of the Indian Association for General Relativity and Gravitation (IAGRG)** organized by the Department of Physics, Indian Institute of Technology Gandhinagar, India.

December 10–13, 2019: Poster Presentation on *"Hawking emission of charged particles from an electrically charged spherical black hole with scalar hair"* at the **9th International Conference on Gravitation & Cosmology (ICGC)** hosted by the Indian Institute of Science Education and Research Mohali, India.

February 23–25, 2019: Oral Presentation on *"Quasinormal modes and superradiant stability of a charged spherical black hole with scalar hair"* in **DPS@10 : The Conference to celebrate 10 years of DPS** organized by Department of Physical Sciences, Indian Institute of Science Education and Research Kolkata, India.

January 3–5, 2019: Poster Presentation *"On the quasinormal modes of a scalar-hairy-Reissner-Nordström black hole"* at the **30th meeting of the Indian Association for General Relativity and Gravitation (IAGRG)** organized by the Department of Physics, Birla Institute of Technology and Science - Pilani, Hyderabad, India.

December 3–7, 2018: Oral Presentation on *"Quasinormal modes of a charged spherical black hole with scalar hair for scalar and*

Dirac perturbations" in the **International Conference on Current Developments in Quantum Field Theory and Gravity** organized by the S. N. Bose National Centre For Basic Sciences, Kolkata, India.

Academic Visits

April 1–5, 2026: Inter University Centre for Astronomy and Astrophysics, Pune

December, 2025: School of Physical Sciences, Indian Association for the Cultivation of Sciences. Hosted by Prof. Sumanta Chakraborty

July 15, 2025: School of Physics and Astronomy, University of St. Andrews. Hosted by Prof. Keith Horne and Prof. Hongsheng Zhao

June 10–26, 2025: School of Physics, Indian Association for the Cultivation of Science Kolkata. Hosted by Prof. Sumanta Chakraborty

February 5–10, 2024: National Institute of Technology, Rourkela, India (NIT-R). Hosted by Prof. Indrani Banerjee

January 1–31, 2024: Inter-University Centre for Astronomy and Astrophysics (IUCAA)

July 1–14, 2023: Physics Department, Indian Institute of Technology Gandhinagar. Hosted by Prof. Sudipta Sarkar

Outreach Activities

31 January 2026: Weekend Lecture on *Black holes as cosmic laboratories for fundamental physics* at the Scientific and Critical Research Institute, Kolkata.

Teaching Experience

Advanced Level

Instructor for 'Blackhole Perturbation theory' course **IUCAA Pune**
, Delivered a set of lectures on Blackhole perturbation theory at the school *October 5 –16, 2026*
"Advanced Topics in General Relativity for Multimessenger Observations"

The course was targetted for PhD students, Postdocs and faculty members.

Teaching Assistant for 'Beyond the Horizon: Testing the black hole paradigm' **ICTS Bangalore**
Instructor: Prof. Dimitrios Psaltis *March 2025*
This is an advanced school for graduate students and postdocs to learn from leading researchers about several crucial aspects of the physics of compact objects.

Teaching Assistant for 'Black holes in General Relativity and beyond' **2nd IAGRG School at ICTS Bangalore**
Instructor: Prof Thomas Sotiriou *October 2023*
This is an advanced course organised by the Indian Association for General Relativity and Gravitation (IAGRG) for graduate students working on gravitational physics, cosmology, and related areas.

Undergraduate/ Postraduate level

IIT Bombay
Teaching Assistant for 'UG - First year Physics Lab' (PH117) *April – June 2023*
This is a Lab course for first year undergraduate students.

IIT Bombay
Teaching Assistant for 'UG - First year Physics Lab' (PH117) *November 2022 – Februaury 2023*
This is a Lab course for first year undergraduate students.

IIT Bombay
Teaching Assistant for 'Basics of Electricity & Magnetism' (PH108) *March 2022 – July 2022*
This is a introductory course on 'Electricity & Magnetism' for undergraduate students.

IISER Kolkata
Teaching Assistant for 'Basic Statistical Mechanics' (PH3201) *January – May 2019*
Course Instructor: Prof. Narayan Banerjee
This is a 4 credit (level 6) introductory course on 'Statistical Mechanics' for undergraduate students.

IISER Kolkata
Teaching Assistant for 'Basics of Field Theory and Relativistic Quantum Mechanics' (PH4106) *August – December 2018*
Course Instructor: Prof. Narayan Banerjee
This is a 4 credit (level 7) introductory course on 'Field Theory' for undergraduate students.

IISER Kolkata
Teaching Assistant for 'Physics IV' (PH2201) *January – May 2018*

Course Instructor: Dr. Siddharth Lal
This is a 3 credit (level 4) intermediate theory course in physics for undergraduate students.

IISER Kolkata
Teaching Assistant for 'Intermediate Electricity and Magnetism' (PH3202) *January – May 2016*
Course Instructor: Prof. Narayan Banerjee
This is a 3 credit (level 6) theory course on 'Electricity and Magnetism' for undergraduate students.

IISER Kolkata
Teaching Assistant for 'Intermediate Quantum Mechanics' (PH3102) *August – December 2015*
Course Instructor: Prof. Supratim Sengupta
This is a 3 credit (level 5) intermediate theory course on 'Quantum Mechanics' for undergraduate students.

Student Mentoring

2022–2024: Saraswati Devi, doctoral candidate, Indian Institute of Technology Guwahati, India. Current Position: Assistant Professor, Department of Physics, Mangaldai College, Tezpur, Assam

2023–2024: Mr. Surajit Ghosh, Master candidate at Indian Association for the Cultivation of Science. Current Position: Graduate student at Wake Forest University, USA

Schools / Workshops/ Conferences Attended

April 20 – 24, 2026: **HEP Software Foundation India and Inter University Centre for Astronomy and Astrophysics (IUCAA) joint computing workshop on Gravitational wave data pipeline at IUCAA, Pune.**

February 10–14, 2025: **Second School on Black Holes and Gravitational Waves** organized by Centre for Strings, Gravitation and Cosmology at Indian Institute of Technology Madras, India.

October 09–20, 2023: **Indian Association for General Relativity and Gravitation (IAGRG) school on Gravitation and Cosmology** organized at International Centre for Theoretical Sciences, Bangalore, India.

May 15–17, 2023: **Gravitational Wave Open Data workshop** organized by the Gravitational Wave Open Science Center.

January 17–22, 2022: **School on Black Holes and Gravitational Waves** organized by the Centre for Strings, Gravitation and Cosmology (CSGC) at the Indian Institute of Technology Madras (IITM), Chennai, India.

January 9, 2020: **One Day Conference on "Observational Aspects of Black Hole Physics"** organized by School of Mathematical & Computational Sciences and School of Physical Sciences, Indian Association for the Cultivation of Science, Kolkata, India.

February 25–27, 2019: **Three Day Conference on "Gravity at Different Length Scales"** organized by the School of Physical Sciences, Indian Association for the Cultivation of Science, Kolkata, India.

March 25, 2017: **7th Topical Conference on Gravity, Cosmology and Astrophysics - Eastern Region (TCGCA - ER)** hosted by the Indian Association for the Cultivation of Science, Kolkata, India.

June 27–July 8, 2016: **Gian Lecture Series on The Black Hole Information Loss Paradox** by Prof. Samir D. Mathur (Ohio State University, USA) organized by the Indian Institute of Technology Gandhinagar, India.

March 19, 2016: **5th Topical Conference on Gravity, Cosmology, Astronomy and Astrophysics - Eastern Region (TCGCA - ER)** organized by the Indian Statistical Institute, Kolkata, India.

September 19, 2015: **4th Topical Conference on Gravity, Cosmology, Astronomy and Astrophysics - Eastern Region (TCGCA - ER)** organized by Department of Physical Sciences, Indian Institute of Science Education and Research Kolkata, India.

Management Skills

November 4, 2017: Member of the organizing committee for the one day lecture series on **Current Trends in Modern Physics - Third Edition** at the Department of Physical Sciences, Indian Institute of Science Education and Research Kolkata, India.

April 1, 2017: Member of the organizing committee for the one day lecture series on **Current Trends in Modern Physics - Second Edition** at the Department of Physical Sciences, Indian Institute of Science Education and Research Kolkata, India.

November 5, 2016: Member of the organizing committee for the one day lecture series on **Current Trends in Modern Physics - First Edition** at the Department of Physical Sciences, Indian Institute of Science Education and Research Kolkata, India.

September 19, 2015: Member of the local organizing committee for the **4th Topical Conference on Gravity, Cosmology,**

Astronomy and Astrophysics - Eastern Region (TCGCA - ER) at Indian Institute of Science Education and Research Kolkata, India.

Computing skills

Programming Language: Fortran, Python [numpy, sympy, bilby(working knowledge), Dynesty (sampler)], C, Java
Scientific Software: Mathematica (xAct), Matlab

Languages

English: Proficient *Schooling*
Bengali: Profocient *Native*
Hindi: Fluent

Complete list of referees

- Narayan Banerjee
  Professor (Retd.),
  Department of Physical Sciences,
  Indian Institute of Science Education and Research Kolkata,
  Mohanpur, West Bengal, India
  ✉ narayan@iiserkol.ac.in ✉ banerjee.narayan@gmail.com

- Sayan Chakrabarti
  Associate Professor,
  Department of Physics,
  Indian Institute of Technology Guwahati,
  Guwahati, Assam, India
  ✉ sayan.chakrabarti@iitg.ac.in ✉ sayan.sinp@gmail.com

- Sumanta Chakraborty
  Associate Professor
  School of Physical Sciences,
  Indian Association for the Cultivation of Science,
  Kolkata, West Bengal, India
  ✉ tpsc@iacs.res.in ✉ sumantac.physics@gmail.com

- S. Shankaranarayanan
  Professor,
  Department of Physics,
  Indian Institute of Technology Bombay,
  Mumbai, Maharashtra, India
  ✉ shanki@iitb.ac.in ✉ shanki.subra@gmail.com

- Santabrata Das
  Professor,
  Department of Physics,
  Indian Institute of Technology Guwahati,
  Guwahati, Assam, India
  ✉ sbdas@iitg.ac.in

- Kazuharu Bamba
  Professor
  Faculty of Symbiotic Systems Science,
  Fukushima University,
  Fukushima 960-1296, Japan
  ✉ bamba@sss.fukushima-u.ac.jp

- Sanved Kolekar
  Associate Professor,
  Indian Institute of Astrophysics,
  Bangalore, Karnatake, India
  ✉ sanved.kolekar@iiap.res.in